Designing a Universal Information Language

Oscar Bender-Stone

March 26, 2026

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Agenda

Introduction

Language Overview

Syntax

Metatheory

Conclusion

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Background

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[1]Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
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- Information helps us **store** and **organize** data

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  - Throughout *all* history: so much data!
  - How to actually use it?
- Inquiry: what *is* information?

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    - Can be a *process*, or *knowledge*

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- Most useful as *thing*

- **Floridi:** Information is **truthful**[1]

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Problems

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- How to *actually* manage information?
  - **Information base:** a database of information
- **Depth:** domain specific data
  - Terminology, kinds of data, etc.
- **Breadth:** wide variety of domains

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  - **Information base:** a database of information
- **Depth:** domain specific data
  - Terminology, kinds of data, etc.
- **Breadth:** wide variety of domains
  - Different principles, methods
  - Example: Physics ≠ Literature!
- **Standardization:** ambiguous interpretations
  - Related issue: being **interoperable**
    - **Interoperable:** a portable data format
    - Example: U.S. Health records
    - Transfer is hard, even for the *same* patient

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Existing Solutions

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[1]Hitzler et al. (2009). *OWL 2 Web Ontology Language Primer*. W3C.
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[1]Lenat (1995). *CYC: A large-scale investment in knowledge infrastructure*. CACM 38(11).

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Existing Solutions

- **Resource Description Language (RDF):**[1]
  - ▸ Used in Web Ontology Language **(OWL)**
  - \+ Built to be **interoperable**
  - — No built-in logic engine
- **Labeled Property Graphs (LPGs):**[2]
  - ▸ Stores **properties** (node/edge metadata)
  - \+ Efficient queries
  - — No logic engine. Hacky tooling/scripts
- **Cyc:**[3]

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- Largest **knowledge base** to date
+ Sophisticated logic engine
— Hard-coded, proprietary rules

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Proposed Solution

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- **Welkin**: an information language
  - Pronounced: “VEL-KIN”

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- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”[1]

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Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”[1]
  - Represents *expansive* nature of language

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Goals

- **Universality:** be completely expressive

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  - Can add optimizations *in* Welkin

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- **Standardization:** be completely *unambiguous*

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  - Be efficient
  - Can add optimizations *in* Welkin
- **Standardization:** be completely *unambiguous* (**In progress**)
  - Have a formal specification

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General Requirements

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General Requirements

1. Express any computer program

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1. Express any computer program
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General Requirements

1. Express any computer program
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   - Established notion: **Turing machines**[1]
2. Enable user-defined meaning

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   - Need effective operations
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2. Enable user-defined meaning
   - Avoid meaningless symbols!
   - Symbol Grounding Problem[2]
   - Treat as free parameter in theory

- Achieve both via **representations**

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- Centered around *representing* things

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- $a - c \to b$: a **represents** b **in context** c

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- $a - c \to b$: a **represents** b **in context** c
  - a: **sign**
  - b: **referent**
  - c: **context**
- Track *faithfulness* of representations via **truth**

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Units

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- Units are **finite programs**
- Built from **Handles**

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  - Name origin: file/resource handles (programs)

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- Each unit contains its own **context** of representations
  - Written with braces $\{a, b, \ldots, z\}$

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  - Representations are *also* units

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Information

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- Major result: can express *any* "reliable" proof

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- Major result: can express *any* "reliable" proof
  - More details later! (Metatheory)

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Rules

- Three categories:

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  - **Primary:**

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    - Transitivity in a context

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Turing Completeness

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- **Theorem:** *every unit corresponds to some program, and vice versa.*

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- Later:

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Turing Completeness

- Part of Universality
- **Theorem:** *every unit corresponds to some program, and vice versa.*
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- Later: completeness for **proofs**!

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- Two main parts

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  - Encoding: characters

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    - Building blocks of strings

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Syntax

- Two main parts
  - Encoding: characters
    - Building blocks of strings
  - Grammar: strings allowed by language

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Encoding

- Characters written in `type-writer font`[1]
- Define US-ASCII: Printable (codes: 32-126) + `EOF` (code: 127)
- Character classes: refer to thesis

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[1]Font used: Intel One Mono (https://github.com/intel/intel-one-mono), licensed under OFL 1.1 (https://github.com/intel/intel-one-mono/blob/main/OFL.txt).

US-ASCII

Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph
32	20	Space	46	2E	.	60	3C	<	74	4A	J	88	58	X	102	66	f	116	74	t
33	21	!	47	2F	/	61	3D	=	75	4B	K	89	59	Y	103	67	g	117	75	u
34	22	"	48	30	0	62	3E	>	76	4C	L	90	5A	Z	104	68	h	118	76	v
35	23	#	49	31	1	63	3F	?	77	4D	M	91	5B	[	105	69	i	119	77	w
36	24	$	50	32	2	64	40	@	78	4E	N	92	5C	\	106	6A	j	120	78	x
37	25	%	51	33	3	65	41	A	79	4F	O	93	5D	]	107	6B	k	121	79	y
38	26	&	52	34	4	66	42	B	80	50	P	94	5E	^	108	6C	l	122	7A	z
39	27	'	53	35	5	67	43	C	81	51	Q	95	5F	_	109	6D	m	123	7B	{
40	28	(	54	36	6	68	44	D	82	52	R	96	60	`	110	6E	n	124	7C	|
41	29	)	55	37	7	69	45	E	83	53	S	97	61	a	111	6F	o	125	7D	}
42	2A	*	56	38	8	70	46	F	84	54	T	98	62	b	112	70	p	126	7E	~
43	2B	+	57	39	9	71	47	G	85	55	U	99	63	c	113	71	q			
44	2C	,	58	3A	:	72	48	H	86	56	V	100	64	d	114	72	r			
45	2D	-	59	3B	;	73	49	I	87	57	W	101	65	e	115	73	s			

Table 1: Printable US-ASCII codes and glyphs.

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Grammar

- Build on two papers
  - First: **Invertible Syntax Descriptions**[1]
    - Processes strings, forwards and back
  - Second: Edelmann et. al.[2]
    - Used in proof of unambiguity
    - Also ensures efficiency
  - More details in thesis

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[1] Rendel & Ostermann (2010). Invertible syntax descriptions. Haskell '10.
[2] Edelmann et al. (2020). *Zippy LL(1) parsing with derivatives*. PLDI 2020.

Grammar

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```
start := units - lexeme -> EOF,
units := "" | {unit - lexeme -> *{"",
"," - lexeme -> units}},
unit := node - lexeme -> *{
  ":=" - lexeme -> choices,
  *{"", arc - lexeme -> chain}
},
choices := *{"", "|"} - lexeme ->
choices_list,
choices_list := *{"",
  chain - lexeme -> *{"", "|" - lexeme
-> choice_list}
},
chain := *{"", "*"} - seq -> path -
lexeme -> *{"", arc - lexeme ->
chain},
arc := right_arc | other_arc,
right_arc := "-" - seq -> node - seq -
> "->",
other_arc := "<-" - seq -> node - seq
-> *{"-", "->"},
node := *{"", "*"} - seq -> path,
path := *{"", modifiers} - seq ->
trailer - seq -> *{"", graph},
modifiers := *{
  "@",
  "~" - seq -> *{"", "@"}
},
trailer := segment - seq -> *{"", "."
- seq -> trailer},
segment := HANDLE - lexeme -> *{"",
graph},
graph :=  "{" - lexeme -> units -
lexeme -> "}",
HANDLE := ID | STRING,
STRING := SQ_STRING | DQ_STRING,
SQ_STRING := "'" - seq -> *{SQ_CHAR,
"\'"} - seq_many_till -> "'",
DQ_STRING := '"' - seq -> *{DQ_CHAR,
'\"'} - seq_many_till -> '"',
ID := ID_CHAR - seq_many_till ->
*{RESERVED, WHITESPACE}
```

Listing 1: Full Welkin grammar.

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Outline of Unambiguity

- Simple grammar class: LL(1)

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[1]Consult: Aho et al. (2006). *Compilers: Principles, Techniques, and Tools* (2nd ed.).

Outline of Unambiguity

- Simple grammar class: LL(1)
- Desirable properties:[1]
  - Unambiguous: every valid input produces *one* output

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Outline of Unambiguity

- Simple grammar class: LL(1)
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  - Efficient parsing
- For our parsers, need Edelmann et. alpause
  - Involves recursive equations on sets

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- Terminology
  - **RE:** accepted by some Turing machine
    - Might not halt!
  - **Formal system:**
    - A decidable set of **formulas**

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    - A computable **verifier** on **derivations**

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    - A computable **verifier** on **derivations**
  - **First Order Logic:**
    - Propositional logic (and, or, not)

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    - A computable **verifier** on **derivations**
  - **First Order Logic:**
    - Propositional logic (and, or, not)
    - Quantifiers (for all, exists)

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Primer

- Terminology
  - **RE:** accepted by some Turing machine
    - Might not halt!
  - **Formal system:**
    - A decidable set of **formulas**
    - A RE set of **rules**
    - A computable **verifier** on **derivations**
  - **First Order Logic:**
    - Propositional logic (and, or, not)
    - Quantifiers (for all, exists)
  - **Peano Arithmetic** (PA): natural numbers + induction
  - **Consistency:** no proofs contain falsehood ($\perp$)

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Formal System Limits

- Many theorems on limit:s

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  - Problem: want to identify + trust proofs!

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Artëmov's Selector Proofs

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Serial-Soundness

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Serial-Soundness

- Extend Artëmov to **serial-soundness:**
  - Secret: *same* technique!
  - Use **partial truh predicates**[1]
    - Approximates truth up to n derviations
- **Serial soundness:** a first-order RE theory T with a selector s.t.:
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- **Self-verifying**: *in* PA, T can prove its own serial-soundness

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- PA $\vdash_T \varphi$ if $T \vdash \varphi$ and T self-verifying

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Reliable Proofs

- How to generalize to *any* theory?

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  - p_φ: S-proof of φ

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Proof Completeness

- **Note:** still has incompleteness!

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  - Basic idea: ZFC is self-verifying!
    - Again, same proof from Artëmov!
  - HUGE leap; how to extend it?
  - Answer: comprehension!

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Crash Course: Recursive Ordinals

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Crash Course: Recursive Ordinals

- Roughly: certain transitive sets ($x \in y \in z \Rightarrow x \in z$)

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    - Adds a **limit case**
- Limit: **Church-Kleene Ordinal** (ω_1^{CK})

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- Limit: **Church-Kleene Ordinal** (ω_1^{CK})
- **Proof-theoretic ordinal:** measures theory strength
  - Important: this is recursive if T is RE *and* sound
  - Bounded *strictly* below ω_1^{CK}!
  - Using inducitve step: can reach Feferman-Schütte ordinal Γ_0[1]
  - For us: how to reach *anywhere* below ω_1^{CK}?

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Crash Course: Simpson Comprehension

- Enter Simpson[1]

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[1]Simpson (2009). Subsystems of Second Order Arithmetic (2nd ed.). Cambridge
[2]Kleene (1944). On the Forms of the Predicates in the Theory of Constructive Ordinals. Am. J. Math. 66(1).

Crash Course: Simpson Comprehension

- Enter Simpson[1]
- Simpson worked on *Reverse Mathematics*
  - Studies the axioms necessary to prove theorems
- We need his comprehension axiom:

$$\exists X.\forall n.(n \in X \Leftrightarrow \Phi(n))$$

  - Φ: very complex formula (Δ^1_1)
    - Also called **hyperarithmetic**
    - By Kleene, this reaches everywhere below ω_1^{CK}![2]

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- ✓ **Constructive**
- TL;DR: Artëmov + Simpson + Reliability = Proofs 💯

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Agenda

Introduction

Language Overview

Syntax

Metatheory

Conclusion

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Comparisons to Existing Solutions

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- **Resource Description Framework:**

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  - Similar ideas/terms about representations

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    - Logical engine

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Comparisons to Existing Solutions (cont')

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Comparisons to Existing Solutions (cont')

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Comparisons to Existing Solutions (cont')

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Comparisons to Existing Solutions (cont')

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  - Welkin:
    - Flexible data format

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Comparisons to Existing Solutions (cont')

- **Cyc:**
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  - Welkin:
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    - Open specification

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Future Work

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Future Work

- Write the Standard

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Future Work

- Write the Standard
- Create a Fast Implementation

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Future Work

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- Query Solvers

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Thank you!

Questions?

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